

NABL Document Intelligence

An evidence-grounded extraction system that reads a laboratory's accreditation paperwork and fills the NABL application form it belongs to — without ever writing a value it cannot point back to a page.

FORMS SUPPORTED 9 151–159	WORKFLOW NODES 16 / 5 use an LLM	OFFLINE TESTS 83 graph · 285 total	COST / 200 PAGES \$0.12
MAX DOCUMENT 2,000 pp	ACCURACY FIGURE Not yet measured		

01 Overview of doc

This report describes the NABL Document Intelligence system as it stands after the September 2026 rewrite, in which the extraction path was replaced by a multi-agent LangGraph workflow. It covers what the system is for, what it must do, the three designs considered and why the agentic one was chosen, what that choice costs, and where it is still weak.

It is written for the engineers who will extend the pipeline and the reviewers who will decide whether to put it in front of a real applicant laboratory. Every figure quoted is either measured against this repository's own corpus or read from a published AWS rate; where a number is an estimate, it says so.

OPEN

No production accuracy figure exists for the agentic pipeline yet. Bedrock currently returns `ValidationException: Operation not allowed` for every model on the configured AWS account — an entitlement problem, not a code defect — so the workflow runs on Gemini today. Benchmarking against the labelled corpus is the gating task before any accuracy claim is made.

02 Goal

Turn a pile of a laboratory's existing paperwork into a substantially complete, auditable NABL application form — and make every filled value traceable to the page it came from.

The measure of success is not "how many fields did it fill" but "how much reviewer time did it save without introducing a value nobody can defend to an assessor." A confidently wrong field is worse than an empty one, because an empty field is obviously the reviewer's job and a wrong one is not.

EXPLICIT NON-GOALS

- Submitting to NABL. The system produces a form; a human files it.
- Judging whether a laboratory *deserves* accreditation. The eligibility wizard checks stated prerequisites, it does not assess competence.
- Modelling NABL 100B. It is a procedure document with no applicant fields — useful as terminology, not as a form.

03 Requirements

3a Functional

#	REQUIREMENT	STATUS
F1	Gate document upload behind an eligibility wizard covering the mandatory prerequisites (standard implemented ≥3–6 months, internal audit and management review complete, valid PT/ILC, dedicated Quality Manager).	BUILT
F2	Accept PDF, DOCX, JPG, PNG, TIFF up to 200 MB and 2,000 pages.	BUILT
F3	Read every page: text layer where one exists, OCR where it does not, and a record for blank and unreadable pages too.	BUILT
F4	Discover field–value pairs dynamically rather than against a fixed key list, and attach an exact source quote to each.	BUILT
F5	Detect duplicates and contradictions, adjudicate them, and preserve every candidate rather than discarding the losers.	BUILT
F6	Map surviving values onto the target form schema by meaning, discovering that schema at runtime.	BUILT
F7	Support all nine real NABL application forms — 151, 152, 153, 153A, 154, 155, 157, 158, 159 — across their two genuinely different shapes.	BUILT
F8	Present a split-pane review UI with the source on the left, the form on the right, and low-confidence fields flagged for confirmation.	DEGRADED
F9	Return exactly one of six terminal verdicts, with reasoning.	BUILT
F10	Resume a crashed run from the last completed node without reprocessing finished chunks.	BUILT

F8 is marked degraded rather than built: the graph works from page text, so a field carries a page number but no rectangle. The reviewer gets a page-level highlight instead of a tight box. See [10c](#).

3b Non-functional

#	REQUIREMENT	TARGET
N1	Auditability — every extracted occurrence persisted with page, chunk, quote and the node that produced it	6 tables, no cascade delete
N2	Determinism — a replayed checkpoint takes the path it took the first time	No clock, no randomness in routers
N3	Bounded cost per document	3 independent loop guards
N4	Bounded wall clock	3,600 s hard timeout
N5	Testability without a network call	83 offline graph tests
N6	No document content in logs	IDs, counts, error types only
N7	Container runs unprivileged	UID 10001
N8	Provider independence — no single LLM vendor is a single point of failure	Ordered chain, 3 links
N9	Survive partial failure — one bad chunk must not lose the other seventy-nine	Per-chunk isolation

04 Background

NABL accreditation paperwork is unusually hostile to naive extraction. A single application draws on GST and CIN proofs, equipment calibration certificates, reference-material certificates, staff CVs, authorised-signatory declarations, PT/ILC reports and a quality manual — each a different document from a different issuer in a different layout, and many of them scans of paper. The form they feed is 15–30 pages of tables whose first several pages are NABL's own amendment sheet and contents, not applicant data. The system reached its current shape through two earlier designs, both of which failed in instructive ways.

REVISION 0 — SCHEMA-GUIDED EXTRACTION

Classify the document into one of seventeen types, look up a fixed field list for that type, ask the model for exactly those fields. Fast and cheap, and it worked on narrow single-purpose certificates. It failed on anything it had no schema for: a quality manual or an unfamiliar report yielded nothing at all, because there was no field list to ask about.

REVISION 1 — UNIFIED TEXT-PLUS-IMAGE EXTRACTION

Send every page's text *and* a raster of every page in a single call, and ask for open-ended `snake_case` keys plus a results table. This fixed the blind spot — the model could describe a document nobody had schematised — and pairing the image with the OCR text measurably stopped a hallucination class where redacted fields were confidently invented.

It broke on length. A single call carrying every page image runs into Bedrock's 25 MB request ceiling at roughly sixty-five pages; a 200-page document at the configured 110 DPI would have been about 76 MB. Worse, the output cap was the real binding limit: a dense report with 269 test rows truncated its JSON mid-object, which meant *nothing* parsed and the whole document failed. There was also no way to say which pages had actually been read.

REVISION 2 — THE AGENTIC WORKFLOW

The current design accepts more calls in exchange for guarantees. Work is split into token-safe chunks with audited page coverage; each chunk is extracted, validated against the real page text, and retried with a *changed* strategy on failure; conflicts are adjudicated rather than silently resolved; and the run ends in an explicit verdict backed by a re-derived quality check. Eleven of its sixteen nodes contain no model call at all.

05 Who's your customer

CUSTOMER	WHAT THEY ARE TRYING TO DO	WHAT THE SYSTEM OWES THEM
Laboratory Quality Manager PRIMARY	Assemble a first-time or renewal NABL application from paperwork already scattered across the lab, without spending weeks retyping it.	A form that is mostly filled, honestly flagged where it is unsure, and never silently wrong.
Internal reviewer	Check the draft before it leaves the organisation.	A queue of exactly the fields that need a human, each with the quote and page that justify it.
NABL assessor	Verify that a submitted claim matches the underlying record.	An audit trail that outlives the process that produced it — which is why deleting an upload keeps its audit row.
Our own engineers	Diagnose a bad extraction.	An answer to “was this a model problem or a logic problem” — the reason the deterministic and semantic work are separated by node.

The Quality Manager is the customer whose time the system exists to save, but the assessor is the customer whose scrutiny sets the bar. Design decisions that trade the second for the first — guessing a plausible value to raise the fill rate — are the ones this project most consistently refuses.

06 Tenets

- Evidence or nothing.** A value that cannot be matched back to text on a real page does not reach the form. There is deliberately no API path that injects an arbitrary value, because that would be a hole straight through the guarantee.
- Preserve, never discard.** Normalisation adds beside the original; merging records a merge rather than deleting occurrences; conflict resolution marks a preference and keeps every candidate. Rejected and hallucinated values are flagged and kept so a reviewer can see what was refused and why.
- Spend the model only on judgement.** Anything decidable by counting pages, comparing strings or checking a schema is decided in Python.

- 4. **Report honestly rather than guess.** Where accuracy conflicts with cost or speed, spend more and tell the truth about what happened. A failed chunk appears in the result; an exhausted budget records why it stopped.
- 5. **Recovery must change something.** Re-sending an identical prompt to a provider that just failed on it is the least likely thing to work.
- 6. **A validation gate never routes forward on failure.** Enforced by the graph's structure, not by convention.

07 Design

7a Challenge

The problem is not extraction accuracy in isolation. It is holding five properties at once, each of which is easy alone and awkward in combination:

- 1. **Unbounded length.** A filled application runs 15–30 pages; supporting bundles run to hundreds. Any design with a fixed per-document call budget silently truncates a real document, and a confident answer drawn from a fraction of the input is worse than a slow one.
- 2. **Unknown schema on both ends.** The source document's fields are not known in advance, and the target form's shape varies across nine variants in two base shapes.
- 3. **Output tokens are the binding limit, not context.** A chunk small enough to *read* is not necessarily small enough to *answer*, and a truncated reply loses the whole chunk's fields rather than just the overflow.
- 4. **Contradiction is normal.** The same serial number appears on a calibration certificate and an equipment table with different formatting; the same address appears three ways. Picking one silently destroys the disagreement an assessor needs to see.
- 5. **The output is a compliance artifact.** “The model said so” is not a defensible provenance.

7b Types of designs considered

Option A – Single-shot multimodal call REJECTED

Send the whole document – every page's text and every page image – in one request and take the JSON back. This was Revision 1. It is the least code and the fewest calls, and it genuinely produced the best results on short documents, because the model sees the entire context at once and can resolve a label on page two against a value on page nine.

Option B – Fixed schema plus per-section retrieval REJECTED

Introspect the target Pydantic schema, group its fields by section, embed the document's pages, and for each section retrieve only the pages that section needs. One call per section rather than per page. This is the cheapest design that scales, and it was partially built – the retrieval and chunking modules survive in the repository.

Option C – Multi-agent workflow with typed state CHOSEN

Sixteen nodes over one typed state object, checkpointed after every node. Five nodes carry a model call – structure analysis, extraction, name equivalence, conflict adjudication, form correspondence. The other eleven are deterministic Python: validation, preprocessing, chunking, response checking, recovery, evidence checking, completeness, form filling, quality control, decision, persistence.

Three independent loop guards bound the work – three extraction rounds, two re-analysis rounds, and three attempts on any single chunk, that last one shared across both loops so a chunk cannot ping-pong between them. A LangGraph recursion limit of 120 is the backstop that fails loudly rather than running forever.

AGENTIC WORKFLOW – EIGHTEEN STEPS IN SEVEN PHASES

Document ingestion	1 User / frontend Upload a document – PDF, image or DOCX	2 FastAPI backend Receive the request, validate the file
	3 Store document Stream to storage with the size cap enforced during the write	
Document processing	4 Processing orchestrator Start the pipeline; checkpoint after every node	5 Preprocessing agent Page-wise text; OCR where there is no text layer
	6 Document analysis agent Sections, tables, repeating entities, candidate fields	7 Chunk manager Token-safe chunks with audited page coverage
Information extraction	8 Dynamic form-mapping agent Discover the target schema; map fields by meaning	9 Extraction agent Each chunk through Nova, JSON out, evidence attached
Completeness check	10 Completeness agent Page, chunk and evidence coverage; information-loss detection	
	11 Missing information? → Targeted re-analysis agent re-processes only the nominated chunks LOOPS TO STEP 9 · MAX 2 ROUNDS	
Conflict resolution	12 Conflict resolution agent Detect duplicates and contradictions deterministically	
	13 Duplicate or conflict? → Adjudicate and mark a preference – every candidate is kept CAPPED AT 40 PER DOCUMENT	

Validation	14 Validation agent Schema, business rules, field completeness — and every value checked against real page text	
	15 Final output valid? → Correction and reprocessing — recovery changes strategy, it does not retry identically MAX 3 ATTEMPTS PER CHUNK	
Output & storage	16 Generate final JSON and form Structured output, every value carrying its evidence	17 Store in database Results, audit log and metrics across six tables
	18 Return result to frontend Extracted values, verdict and the manual-review queue	

Three loop guards bound the two cycles above — three extraction rounds, two re-analysis rounds, and three attempts on any single chunk, the last shared across both so a chunk cannot ping-pong between them. The node-level compiled graph is available at `GET /graph/diagram`.

WHY THE MODEL IS SPENT WHERE IT IS

NODE	LLM	RATIONALE
Document analysis	Yes	Finding sections, tables and repeating entities is genuine structural judgement.
Dynamic extraction	Yes	The core task — discovering field–value pairs with no key list.
Normalisation & merging	Yes	Deciding that “Lab Name” and “Name of Laboratory” are the same field is semantic — but a deterministic veto blocks merges whose members disagree on data type.
Conflict resolution	Yes	Detection is deterministic; only adjudication is semantic.
Form mapping	Yes	Meaning-level correspondence — but exact name matches are settled in code first, before asking.
Evidence validation	No	String comparison against real page text. Never delegate the integrity check to the thing being checked.
Completeness	No	Counting page and chunk coverage.
Quality control	No	Re-derives every claim from state rather than reading the flags earlier nodes set.

7c Pros

OPTION C — WHAT IT BUYS • Scales with the document, not a budget. Eighty chunks if a document needs eighty chunks. No silent truncation. • Correctness is unit-testable. Eleven deterministic nodes mean 83 tests run with no network call, and a defect has an address. • Resumable. The thread ID is the document ID; a crash at chunk 41 of 80 resumes at 41. • Partial failure is survivable. One bad chunk is reported, not fatal. • Recovery is adaptive — split an oversized chunk, re-OCR at 2× DPI on poor text, back off on a rate limit. Different cause, different action. • The audit trail is the product. Per-occurrence rows answer “how many times did this document state this, and where”. • Cost falls per page as documents grow — the four fixed calls amortise.
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WHAT THE REJECTED OPTIONS GOT RIGHT • A gives the model whole-document context, so a label and its value pages apart resolve naturally. C loses this at chunk boundaries and compensates with a one-page overlap. • A sends the page image, which measurably suppressed a redacted-field hallucination. C is text-only. • A is one call and almost no orchestration code. • B is the cheapest scaling design, and its retrieval index makes targeted re-analysis natural. • B numbers repeating entities correctly within one call, with no cross-call re-indexing.

7d Cons

COST OF CHOOSING C • Far more code. Sixteen nodes, a 30-type error taxonomy, six tables and a checkpointier, against one function in Option A. • More calls. 22 for a 200-page document against A's one — though A cannot process 200 pages at all. • Text-only extraction. Dropping the page image gives up the cross-check that fixed a real hallucination class. This is the most defensible thing to revisit. • No bounding boxes. The graph works from page text, so the review UI degrades from a tight box to a page-level highlight — a direct regression against F8. • More moving parts to misconfigure — chunk size, concurrency, rate limit, three retry ceilings and four thresholds. • Chunk boundaries can split a table across two calls. One page of overlap mitigates; it does not eliminate.
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WHY A AND B WERE REJECTED

- **A hits a hard ceiling.** ~65 pages of images against Bedrock's 25 MB request cap; a 200-page render is ~76 MB.
- **A fails whole-document on truncation.** 269 test rows truncated the JSON mid-object and nothing parsed.
- **A cannot say what it read.** No page coverage, no evidence, no verdict.
- **B only finds what the schema already names** — the exact blind spot Revision 0 was rewritten to fix.
- **B's retrieval can silently miss** the one page holding a section's answer, and nothing detects the miss.
- **Neither A nor B is resumable.** A failure at any point costs the whole document.

08 Conclusion

The agentic design is the right choice for this problem, and the reason is narrower than “agents are better.” The output here is a compliance artifact. A system that fills a form correctly but cannot show its work has not solved the problem, and both simpler designs were structurally unable to show their work — not because they were badly built, but because neither had a place to put a page coverage audit or an evidence check.

The price is honest and affordable: about **\$0.12 per 200-page document**, 22 model calls, and roughly ten times the code of Option A. Cost per page *falls* as documents grow, from \$25.56 per thousand pages on a single-page upload to \$0.58 at 300 pages, because the four fixed calls amortise.

Two things should be revisited rather than accepted. **Text-only extraction** gave up a page-image cross-check that measurably suppressed a hallucination class; re-introducing the image for chunks whose text quality is poor would cost 230 tokens per page and is probably worth it. **Bounding boxes** are a real regression against the review UI's original design, and recovering them means threading PyMuPDF span geometry through chunking and matching evidence back to spans.

The gating task is measurement. Every claim about this pipeline's accuracy is currently unmade. The labelled corpus and the scoring harness both exist; what is missing is a Bedrock entitlement and a benchmark run. Until that lands, the honest statement is that the system is well-built and unproven.

09 FAQs

– Why not just use a bigger context window and send the whole document?

Context is not the binding limit — output tokens are. Nova 2 Lite accepts a million input tokens, so a 200-page document fits comfortably. What does not fit is the *reply*: a document with hundreds of test rows exceeds the output cap, truncates mid-object, and then nothing parses at all. Chunking exists to bound the answer, not the question.

– Sixteen nodes for document extraction — isn't that over-engineered?

It would be, if all sixteen were model calls. Eleven are ordinary Python that could have been written as helper functions; making them nodes is what makes them individually testable, individually checkpointed, and individually visible in the audit log. The genuine complexity cost is the three loop guards and the routing, and that is the price of not truncating long documents.

– Why is a hallucinated value kept rather than deleted?

Because a reviewer needs to see what was rejected and why. A deleted value is indistinguishable from a value that was never extracted, which makes it impossible to tell a cautious run from a blind one. Rejected values are downgraded and flagged; the form-filling node never writes them.

– Which model does this actually run on today?

Gemini. The chain is ordered `nova, gemini, groq` with Nova 2 Lite as the intended primary, but Bedrock currently rejects every model on the configured account with `ValidationException: Operation not allowed` — an account entitlement issue. The chain exists precisely so this is a degradation rather than an outage. All costs in this report are quoted at Nova rates, since that is the intended production path.

– Where does the deployed OCR Lambda fit in?

Narrowly. `akash-ocr` is a container-image Lambda in `ap-south-1` that takes an image and returns text, lines, confidences and bounding boxes — it extracts no fields. It ships recognition models for Devanagari, Arabic, Tamil, Telugu and Kannada only. English, which is the entire current corpus, runs RapidOCR in-process with no AWS call, so on today's documents the deployed Lambda is not in the path at all.

– Only 7.4% of pages need OCR. Why carry an OCR path at all?

Because the 7.4% is not evenly spread — it is concentrated in whole scanned documents, typically the older calibration certificates and signed declarations that carry exactly the values an assessor checks. A born-digital-only system would work on most pages and fail on the ones that matter most.

– What stops the retry logic from looping forever?

Three independent guards, deliberately not nested so one cannot mask another's runaway: three extraction rounds, two re-analysis rounds, and three attempts on any single chunk. The third is shared across both loops, which is what stops a chunk ping-ponging between them. A recursion limit of 120 catches a routing bug that escapes all three, and fails loudly.

– Can a reviewer type in a value the document doesn't contain?

Not through the graph's review endpoint. A decision names a `field_uid` — one of the candidates the document actually contains. There is deliberately no API path that injects an arbitrary value, because that would defeat the evidence guarantee that the whole design exists to provide. Free-form correction remains available on the legacy field-patch endpoint, outside the audit chain.

– Why keep the audit record when a document is deleted?

The audit trail is the compliance artifact — it is the thing an assessor may need years after the source file has been purged under a retention policy. Deleting an upload removes the file and keeps the record, deliberately. Note the corollary in [10c](#): checkpoints hold full page text, so they need pruning separately.

10 Appendix

10a Metrics

CORPUS CHARACTERISTICS – MEASURED, 40 PDFS / 298 PAGES

MEASURE	MEDIAN	MEAN	P90	MAX
Characters per page	1,665	2,122	3,600	7,670
Tokens per page (chars ÷ 4)	416	531	900	1,918
Pages lacking a text layer	22 of 298 – 7.4% require OCR			

EXTRACTION YIELD – MEASURED, 52 CACHED PREDICTIONS

MEASURE	MEDIAN	MEAN	P90	MAX
Output JSON tokens per document	754	773	1,072	2,960
Header fields per document	22	—	—	—
Test-table rows per document	12	—	—	51

CONFIGURED THRESHOLDS

SETTING	VALUE	GOVERNS
MAX_CHUNK_CHARS	24,000	Chunk size — the cheapest cost lever, and the riskiest above ~32,000
max_fields_per_chunk	120	Hard cap on the JSON one call may be asked to produce
verified_confidence_threshold	0.75	Below this a field goes to manual review
evidence_similarity_threshold	0.88	How closely a quote must match real page text
mapping_confidence_threshold	0.70	Below this a form mapping is not applied
low_confidence_ratio_limit	0.30	Fraction of low-confidence fields that trips a QC issue
MAX_CONCURRENT_LLM_CALLS	4	Wall clock on long documents
MAX_REQUESTS_PER_MINUTE	60	Stays inside the provider allowance
OCR_DPI	200	Quality against memory and time; recovery re-OCRs at up to 2×
workflow_timeout_seconds	3,600	Hard wall-clock ceiling per document

TEST COVERAGE

SUITE	TESTS	COVERS
test_nodes_unit.py	50	Each deterministic node in isolation
test_integration.py	10	Full runs, output contract, resumability, human-in-the-loop
test_failures.py	23	Every failure class in the specification
Graph total	83	All offline — every model call scripted
Repository total	285	Including the pre-existing pipeline suite

10b Performance

COST MODEL – NOVA 2 LITE AT \$0.30 / \$2.50 PER MILLION TOKENS

PAGES	CHUNKS	CALLS	INPUT TOK	OUTPUT TOK	COST	PER 1K PAGES
1	1	5	22,700	7,500	\$0.0256	\$25.56
10	1	5	22,700	7,500	\$0.0256	\$2.56
30	3	7	36,100	10,500	\$0.0371	\$1.24
100	9	13	76,300	19,500	\$0.0716	\$0.72
200	18	22	136,600	33,000	\$0.1235	\$0.62
300	27	31	196,900	46,500	\$0.1753	\$0.58

Four fixed calls per document — analysis, merge, conflict, mapping — on top of one per chunk. Chunk count derives from the measured 2,122 characters per page against the 24,000-character chunk limit.

RETRY AND RE-ANALYSIS OVERHEAD, WITHIN THE LOOP GUARDS

SCENARIO	COST / 200 PP	UPLIFT
Clean run	\$0.1235	—
One extra extraction round on 20% of chunks	\$0.1482	+20%
Both re-analysis rounds consumed	\$0.1852	+50%

CHUNK-SIZE SENSITIVITY ON A 200-PAGE DOCUMENT

MAX_CHUNK_CHARS	CHUNKS	CALLS	COST	RISK
8,000	54	58	\$0.2660	Safe replies; 2.2× the cost
16,000	27	31	\$0.1591	Conservative

MAX_CHUNK_CHARS	CHUNKS	CALLS	COST	RISK
24,000	18	22	\$0.1235	Default
32,000	14	18	\$0.1088	Measure before going past this — a truncated reply costs the whole chunk

MONTHLY SPEND AT VOLUME – 200-PAGE DOCUMENTS, CLEAN RUNS

DOCUMENTS / MONTH	MODEL COST
100	\$12.35
1,000	\$123.48
5,000	\$617.40

PREPROCESSING LATENCY – MEASURED ACROSS 278 REAL PHOTOGRAPHS

SIZE BUCKET	PHOTOS	AVERAGE	SLOWEST
Large (to 48 MP)	79	1,248 ms	2,427 ms
Medium	91	896 ms	1,948 ms
Small	108	670 ms	2,012 ms

10c Edge cases

HANDLED

CASE	BEHAVIOUR
Blank page	Gets its own zero-content chunk marked <code>SKIPPED_EMPTY</code> — accounted for, no call spent. An omitted page would otherwise look like a page that never existed.
Page in no chunk	Chunking audits its own coverage and raises <code>PAGE_NOT_CHUNKED</code> as a critical completeness failure.
Oversized page	Split, never dropped. Page claims are preserved across the split.
Truncated JSON reply	Chunk split in half and both halves requeued — the recovery changes something rather than retrying identically.
Poor OCR quality	Chunk's pages re-OCR'd at up to 2× DPI, rebuilt and requeued.
Rate limit / timeout	Requeued unchanged; exponential backoff doubles from 2 s to a 60 s cap <i>with jitter</i> — without jitter, N chunks that fail together retry together and reproduce the burst.
Provider cooldown past 60 s	Waits up to 240 s. Real cooldowns of 78 s and 158 s were measured, so a 60 s ceiling was abandoning chunks that would have succeeded.
Twelve identical equipment rows	Survive deduplication — field identity is (name, page, chunk, occurrence, value, evidence), not name. Identical duplicate <i>replies</i> do not survive.
Conflict resolved on confidence alone	Downgraded to manual review rather than accepted.
Database write fails	Persistence never fails the workflow; the result is not lost.
Password-protected PDF	Refused, not cracked.
<code>.pdf</code> without a <code>%PDF</code> header	Rejected as corrupt — extension allow-list, then content verification.
Oversized upload	Size cap enforced <i>during</i> the streaming write, so memory cannot be exhausted before rejection.
Persistent LLM failure	Terminates with a recorded stop reason rather than looping.

KNOWN LIMITATIONS – STATED RATHER THAN DISCOVERED IN PRODUCTION

LIMITATION	CONSEQUENCE	SEVERITY
No bounding boxes	A field carries a page number but no rectangle; the review UI degrades to a page-level highlight. Fixing it means threading PyMuPDF span geometry through chunking.	PRODUCT
Nova unverified end to end	Bedrock returns <code>ValidationException</code> account-wide; the workflow runs on Gemini. No production accuracy figure exists.	BLOCKING
DOCX pagination is synthetic	No page concept exists, so it is chunked at 3,000 characters per synthetic page. Page numbers are positional, not real.	PRODUCT
Semantic merge skipped above 250 distinct field names	Exact matching still applies; very wide documents get less merging.	ACCEPTED
Conflict adjudication capped at 40 per document	The remainder go straight to manual review rather than being guessed at.	ACCEPTED
Chunk boundary can split a table	One page of overlap mitigates; it does not eliminate.	PRODUCT
Checkpoints hold full page text	For sensitive corpora, prune after completion — the result row is self-contained. PII lives in clear in <code>graph_fields.value</code> and <code>exact_source_evidence</code> ; on Postgres, use column encryption or a restricted role.	PRIVACY
Several 154 and 158 tables are schema-only	Mobile Laboratory, Complaints, FSSAI-scope, Sampling-scope, Affiliations, Family-members, Related-labs and Contracts are modelled and editable but not populated from an upload.	ACCEPTED

Two defects this build found in itself, both surfaced by generating a sample rather than by the test suite — a reminder that a passing suite is not the same as a working system. `clean_text()` collapsed newlines, folding each page into one line and destroying the line structure that table reading and evidence quoting depend on. And the minimum-evidence-length check measured the alphanumeric-only match form, so `Fat: 4.2 %` became `fat42` — five characters — and a perfectly good quote was rejected as too short. That silently invalidated most short labelled numeric fields, which is most of a lab report.

NABL

National Accreditation Board for Testing and Calibration Laboratories — the Indian body that accredits laboratories.

151 – 159

The nine real NABL application forms. 151/152/153/153A/154/158 are full ISO/IEC 17025 or ISO 15189 accreditation applications; 155/157/159 are lighter PT-performance-based recognition schemes with a different shape.

NABL 100B

An accreditation-procedure document, not an application form. Useful as terminology reference; deliberately not modelled.

MU

Measurement Uncertainty — the testing and medical disciplines' expression of doubt about a result. Kept separate from CMC because NABL treats them as distinct concepts.

CMC

Calibration and Measurement Capability — the calibration discipline's counterpart to MU.

Z-score

Proficiency-testing performance metric used in testing and medical scopes.

 E_n -value

The calibration discipline's PT performance metric, the counterpart to a Z-score.

PT / ILC

Proficiency Testing / Inter-Laboratory Comparison — externally run exercises that demonstrate a lab produces correct results. A valid recent result is an eligibility prerequisite.

MRM

Management Review Meeting — a prerequisite the eligibility wizard checks before unlocking upload.

Chunk

A token-safe slice of a document, at most 24,000 characters, preserving page boundaries with one page of overlap. Sized by what the model can *answer*, not what it can read.

Occurrence

One statement of a field in the document. Identity is (name, page, chunk, occurrence, value, evidence) — so twelve equipment rows are twelve fields, not one.

Evidence

The exact source quote backing a value, checked against real page text by a deterministic node. A value without one does not reach the form.

Checkpoint

Persisted state after every node, keyed by document ID, which is what makes a crashed run resumable and targeted re-analysis possible.

Loop guard

One of three independent bounds — extraction rounds, re-analysis rounds, per-chunk attempts — kept separate so one cannot mask another's runaway.

deepreef-ocr

The sibling OCR project, deployed as the `akash-ocr` container Lambda in ap-south-1. Image in, text and boxes out; extracts no fields.

RapidOCR

The ONNX OCR engine behind both paths — baked non-English models in the Lambda, its own bundled English model in-process locally.

Inference profile

The region-prefixed Bedrock model identifier (`us.amazon.nova-2-lite-v1:0`). Nova 2 Lite rejects the bare model ID for on-demand invocation.